- MLX90640 thermal camera
- ❌ Raspberry Pi
- ❌ LiDAR
- ❌ camera-based pest AI
- ❌ LoRa
- ❌ NPK optical spectroscopy
- ❌ stepper motors
- ❌ expensive weather station

Component	Purpose	Approx. cost
ESP32-S3 Dev Board	Main controller + Wi-Fi	â‚¹500â€“700
SHT40/SHT41	Air temperature + humidity	â‚¹300â€“500
MLX90614	Non-contact leaf/canopy temperature	â‚¹600â€“900
Capacitive leaf-wetness sensor/grid	Detect leaf moisture/disease-favourable wetness	â‚¹150â€“300
RS485 NPK/soil sensor	Soil moisture + EC + soil temperature	â‚¹1,500â€“2,000
MAX485 module	ESP32 â†” RS485 communication	â‚¹50â€“100
Rain sensor/gauge	Rain detection	â‚¹100â€“300
Light/PAR sensor	Solar/light intensity	â‚¹150â€“400
12V mini water pump	Demonstrate automatic irrigation	â‚¹250â€“400
1-channel/2-channel relay	ESP32 controls pump	â‚¹50â€“150
Water pipe + small tank	Irrigation demo	â‚¹150â€“300
Solar panel 10W	24-hour power demonstration	â‚¹600â€“900

18650/Li-ion battery + BMS	Night-time power	â‚¹300â€“500
Solar charge controller/ MPPT	Solar charging	â‚¹200â€“400
IP65 junction box	Protect electronics	â‚¹250â€“400
PVC/aluminium tower	Physical structure	â‚¹300â€“600
Wires/connectors/PCB/ breadboard	Assembly	

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